Calculus Proofs

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1 Sample Proof

This document demonstrates the basic LaTeX setup for writing calculus proofs.

Theorem 1.1 (Derivative of a Constant)**.** *If c is a constant, then $\frac{d}{dx}c = 0$.*

Proof. By the definition of derivative:

$$\begin{aligned}
\frac{d}{dx}c &= \lim_{h \to 0} \frac{c - c}{h} && (1)\\
&= \lim_{h \to 0} \frac{0}{h} && (2)\\
&= \lim_{h \to 0} 0 && (3)\\
&= 0 && (4)
\end{aligned}$$

$\square$

Example 1.2. Consider the function $f(x) = 5$. Then $f'(x) = 0$ for all $x \in \mathbb{R}$.

2 Additional Proofs